

Volume 3: The Physics — Index

2026-05-19

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Target reader: Engaged physicist working with the framework; wants full BST physics

Voice/style: Technical, dynamics-level

Chapters

Chapter	Topic	Contents
Ch01_Gravity_Vacuum.md	Gravity and Vacuum	Sections 10-12: Gravity as Statistical Thermodynamics (Newton's G), Chiral Condensate, Vacuum Energy
Ch02_Quantum_Interface.md	Quantum-Classical Interface (Hamiltonian + Bergman Dirac)	Section 13: includes Hamiltonian $H = (\text{winding})^2$, $\hbar = 2mD$, Born Rule from Gleason, 13.9 Bergman Dirac Operator (Spring 2026 LAG-1 Sessions 1-10 full operator-level work, eigentones $\lambda=10 + \lambda=75/4$)
Ch03_Forces_Cosmology.md	Forces and Cosmology	Sections 14-21: Three Geometric Layers, Cosmological Implications, Matter Clumping, Information Theory, 2D-to-3D Interface, Dark Matter as Channel Noise, Weak Force as Variation Operator, Thermodynamic Foundation
Ch04_Cosmic_Cascade.md	Cosmic Architecture	Sections 22-25: Arrow of Time, Wavefront, Growing Manifold, Cascade of Forced Choices
Ch05_Discussion.md	Discussion — Lagrangian Status + Partition Function + Central Claim	Section 26: Six-term Lagrangian S_{BST} status, Partition Function as Master Calculation, QM and GR as scale limits of one substrate, Arrow of Complexity

Supporting computational subdirectory

- `rotation_curves/` — SPARC galaxy rotation curve data + `sparc_bst.py` analysis script + `DarkMatter-Calculation.md` documenting BST's channel-noise framework for galaxy rotation curves (no dark matter particles required). Directly supports Chapter 3 (Dark Matter as Channel Noise, originally Section 19 of monolithic). Moved into Volume 3 on 2026-05-19 from repository root.

Build instructions

Build the volume PDF with pandoc:

```
cd Guide/Vol3_Physics  
pandoc Ch*.md -o Volume.pdf --pdf-engine=xelatex -H ../notes/bst_pdf_header.tex
```

Navigation

- Up: Master Library Index
- All volumes: Vol 1 Journey, Vol 2 Framework, Vol 3 Physics, Vol 4 Mathematics, Vol 5 Predictions, Vol 6 Frontier

— Keeper, 2026-05-19